Lab 7: Implementing and Analyzing Mobile IP

Objective:

By the end of this lab, students will be able to:

1. Understand the concept and architecture of Mobile IP (MIPv4/MIPv6).
2. Configure and simulate a basic Mobile IP scenario.
3. Observe and analyze the behavior of a mobile node moving across different networks.
4. Use Wireshark or other tools to inspect Mobile IP protocol messages (e.g., registration, tunneling).

Lab Requirements:

- Software:
  - GNS3 / Mininet / Cisco Packet Tracer (GNS3 preferred for realism)
  - Ubuntu VMs (Mobile Node, Home Agent, Foreign Agent)◦ Wireshark
  - Optional: NS-3 for advanced students
- Hardware:
  - Lab PCs with virtualization support
  - OR student laptops with VirtualBox / VMware

Lab Topology

[Mobile Node] <==> [Foreign Agent] <==> [Internet] <==> [Home Agent] <==> [Home Network]

Task 1: Introduction and Setup

Install and configure Ubuntu virtual machines or containers for:

- Home Agent (HA)
- Foreign Agent (FA)
- Mobile Node (MN)

Assign IP addresses and ensure network connectivity:

- Home Network: 192.168.1.0/24
- Foreign Network: 192.168.2.0/24

Task 2: Configure Basic IP Routing

- Enable IP forwarding on HA and FA.
- Set up static routing or use a dynamic routing protocol.
- Test ping connectivity between all nodes.

Task 3: Configure Mobile IP

- On the Home Agent, configure:
  - Home Address Binding
  - Tunnel interface (e.g., GRE/IP-in-IP)
- On the Foreign Agent, configure:
  - Care-of Address assignment
  - Forwarding agent
- On the Mobile Node:
  - Assign permanent home address
  - Configure agent discovery and registration

Task 4: Test Mobility

1. Before movement:

MN is on the home network. Ping a correspondent node (e.g., Internet or HA). Observe a stable IP connection.

2. Simulate movement:

◦ MN detaches from the home network and connects to the foreign network (change VM network interface).
◦ MN registers with the Foreign Agent.
◦ HA creates a tunnel to the MN's care-of address.

3. After movement:

◦ Continue pinging to verify session continuity.
◦ Capture and analyze registration messages and tunnel packets using Wireshark.

Task 5: Report and Reflection

• Answer lab questions such as:

◦ What happens to the IP address of the MN after moving?
◦ How does the tunnel work?
◦ What are the advantages/disadvantages of Mobile IP?

Optional Advanced Exercises

- Implement Mobile IPv6 using NS-3.
- Observe handoff latency.
- Compare Proxy Mobile IP (PMIPv6) with MIPv6.
- Simulate real-world scenarios (e.g., VoIP session mobility).

Submission Format:

A single PDF report named `CO513_Lab07.zip`, with all of the screenshots, results, and analysis. Please make sure to include all the group members' enumbers in the report.